

The Opportunity Cost of Concentrating Payroll: How the NBA's Second Apron Reshapes Roster Construction

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Abstract

The NBA's 2023 Collective Bargaining Agreement introduced the second apron, substantially increasing the financial and roster-building constraints faced by teams with the league's highest payrolls. This paper examines how these changes have reshaped payroll allocation and roster composition, and what this implies for team performance, using team- and player-level panel datasets spanning all 30 NBA franchises over 15 seasons. The results suggest that, while average payroll concentration did not increase significantly following its introduction, a larger share of teams adopted concentrated payroll structures. Across the full sample, greater payroll concentration is associated with a smaller mid-level veteran minute share, reflecting fewer rostered mid-level veterans rather than reduced playing time for those retained, while teams allocating more minutes to mid-level veterans won more games after controlling for payroll and star-player quality. Payroll concentration itself has no independent relationship with winning on average once star-player quality is taken into account, but that null result masks a genuine tradeoff: improved talent quality always helps team performance regardless of how payroll is split; however, concentrating further around that talent leaves fewer resources for the rest of the roster, and this cost grows larger as the concentrated talent becomes more elite. A case study of Jaylen Brown and the Boston Celtics — traded by the team in 2026 — illustrates this tradeoff in a real roster-building decision. Overall, the second apron has increased the opportunity cost of concentrating payroll, making successful roster construction increasingly dependent on balancing investment in elite talent with roster depth.

I. Introduction

1.1 Context and Why It Matters

The NBA operates under a soft salary cap, permitting teams to exceed the cap through collectively bargained exceptions, but discouraging excessive spending through a luxury tax that imposes escalating per-dollar penalties above a set threshold — penalties that grow steeper still for teams that repeatedly exceed it. For context, in the 2025-26 season the salary cap sits at \$154.6 million, the luxury tax threshold at \$187.9 million, and the second apron at \$207.8 million; a team at the second apron is spending roughly \$53 million above the cap (NBA.com, 2025).

The NBA's second apron represents one of the most significant changes to roster construction in the league's modern salary cap era. Introduced as part of the 2023 Collective Bargaining Agreement (CBA), the second apron imposed a new layer of roster-building restrictions on top of these existing financial penalties.¹ The new rules restrict access to several key team-building mechanisms — including the taxpayer mid-level exception, salary aggregation in trades, cash considerations, and future draft flexibility (Sportico, 2024) — making it substantially more difficult for high-spending teams to maintain expensive, veteran-heavy rosters. While the first apron had imposed some roster-building restrictions since 2011 (Sheppard Mullin, 2025), the second apron's rules are far more severe, leaving teams unable to simply pay their way past them the way they often could under earlier rules. As a result, teams now face a more explicit tradeoff between star talent and roster depth. Rather than asking whether teams can simply afford to pay elite players, front offices operating under these rules must increasingly evaluate whether the value those players generate outweighs the roster depth sacrificed and the roster-building limitations that come with supporting them.

These changes arrive at a time when payroll concentration has become increasingly common across the league. The rapid growth of maximum contracts, combined with rising player salaries, has led many franchises to devote an increasing share of payroll to a small number of star players. While concentrating payroll may be justified when elite talent produces championship-level value, doing so also limits the financial flexibility available to retain

¹ The NBA's second apron was introduced as part of the 2023 Collective Bargaining Agreement beginning with the 2023–24 season. While several restrictions were phased in over the following two seasons, 2023–24 marks the first season in which the second apron existed as a distinct payroll threshold. Accordingly, this paper defines the post-apron period as 2023–24 onward, while recognizing that only the 2024–25 and 2025–26 seasons were subject to the apron's full set of roster-building restrictions (Hoops Rumors, 2025).

experienced role players and mid-level veterans. Under the second apron, these opportunity costs have become increasingly important, raising a central question for NBA front offices: how should teams allocate payroll to maximize on-court success?

1.2 Research Question

Despite the growing attention devoted to the NBA's second apron, relatively little empirical evidence exists on how teams have adjusted their roster construction in response to these new constraints or whether those adjustments improve competitive outcomes. Existing discussions have largely been descriptive, focusing on the financial implications of the new CBA and the challenges of retaining expensive rosters, while offering limited empirical evidence on how payroll allocation has changed across the league or the competitive consequences of those changes.

This paper asks three related questions. First, have teams become increasingly concentrated in how they allocate payroll among their highest-paid players? Second, how does greater payroll concentration affect roster composition, particularly the role of mid-level veterans? Finally, do these roster-building strategies improve team success, or do they impose costs that outweigh the benefits of concentrating resources among elite talent?

1.3 Summary of Findings

The results suggest that, although the league-wide average level of payroll concentration did not increase significantly following the introduction of the second apron, a significantly larger share of teams adopted relatively concentrated payroll structures. Across the sample, greater payroll concentration is associated with a smaller share of minutes going to mid-level veterans. This decline reflects teams rostering fewer mid-level veterans, rather than simply playing the same veterans fewer minutes. Yet teams that do allocate a larger share of minutes to mid-level veterans consistently win more games after controlling for overall payroll and the quality of their highest-paid players. However, payroll concentration itself has no independent relationship with winning once player quality is taken into account. Rather, concentrating payroll creates a tradeoff between investing in elite talent and maintaining roster depth, a tradeoff illustrated directly by a closing case study of Jaylen Brown and the Boston Celtics.

II. Literature Review

This paper draws on two related strands of research: work directly studying spending above the NBA salary cap, and a broader debate over whether concentrating talent helps or hurts team performance.

The closest existing precedent is Pradhan and Leshchinskii (2025), who study the years before the second apron existed and test whether spending above the salary cap actually translates into the kind of performance gains that justify the resulting luxury tax penalties. At the time of their study, the costs of high spending were financial penalties and the more modest restrictions carried by the original, single-tier apron in place since 2011. The second apron sharply raises the severity of those restrictions, making it considerably harder for high-spending teams to fill out the rest of their roster. This paper builds on that work by asking a more specific question — not simply whether spending and winning are correlated, but how teams can allocate payroll most efficiently within this stricter tax and apron structure, and how front offices are actually navigating these added constraints. This is a question Pradhan and Leshchinskii explicitly set aside, noting the second apron had not yet taken full effect within their sample, and calling directly for future work on its effect on team performance and roster construction.

Beyond this specific precedent, a broader and older literature has directly debated whether concentrating talent in a small number of players helps or hurts team performance. Swaab, Schaerer, Anicich, Ronay, and Galinsky (2014) found evidence of a "too-much-talent effect" using a decade of NBA data: team performance rose with the share of a team's top talent up to a point, then declined, with an estimated turning point around 52% of a team's talent concentrated in its most productive players—a threshold notably close to the top-heavy cutoffs (50–55%) examined in this paper. Gula, Vaci, Alexandrowicz, and Bilalić (2021) later challenged this finding on methodological grounds, reanalyzing the same and a larger dataset and concluding that additional concentrated talent never becomes harmful. Instead, each additional unit of concentrated talent continues to improve team performance, albeit with diminishing marginal returns.

This paper approaches that debate from a different angle. Rather than measuring concentration in talent-weighted terms, it measures concentration in payroll terms and then separately controls for the quality of a team's highest-paid players. This distinction allows the analysis to compare teams with similarly productive stars but different payroll structures, isolating whether allocating a larger share of payroll to those stars — rather than distributing those payroll resources more broadly across the rest of the roster — is itself associated with winning. The results suggest it is not. Once the quality of a team's highest-paid players is accounted for, payroll concentration has no independent relationship with team performance. In other words, any observed relationship between

top-heavy payroll structures and winning is explained by the quality of the players receiving the largest share of payroll, not by payroll concentration as a roster-building strategy in its own right.

These findings also offer a potential explanation for why concentrating resources may exhibit diminishing returns. Greater payroll concentration is separately associated with carrying fewer mid-level veterans, while greater reliance on those veterans is independently associated with winning. Together, these findings suggest that one important consequence of concentrating payroll is reduced roster depth, providing a concrete roster-construction channel through which the returns to concentrating financial resources may diminish. Rather than viewing payroll concentration as inherently beneficial or harmful, this paper reframes it as a financing decision whose success depends on whether the value generated by a team's highest-paid players outweighs the opportunity cost of the roster depth sacrificed elsewhere.

III. Data

A. Data Sources

This study combines complementary team- and player-level panel datasets. The team-level dataset includes all 30 NBA franchises over 15 seasons (2011-12 through 2025-26), producing 450 team-season observations before accounting for lagged specifications. Team payroll information — including taxable salary, luxury tax thresholds, and payroll by player — was collected primarily from Spotrac and HoopsHype, while team performance measures were obtained from Basketball-Reference.

To examine roster composition, I construct a complementary player-level dataset containing 3,645 player-season observations. Player salary information was collected from Spotrac, playing time and box-score statistics from Basketball-Reference, and Estimated Plus-Minus (EPM) data — used in the closing case study — from Dunks & Threes. Combining these datasets connects team-level payroll decisions to changes in player utilization and, ultimately, team performance.

The sample spans multiple collective bargaining agreements and captures the second apron's 2023 introduction, providing substantial variation in payroll allocation strategies before and after the policy change. Because the post-apron period currently includes only three seasons — only two of which faced the apron's full

restrictions — results specific to that era should be read as an early evaluation of teams' initial responses rather than a long-run estimate.

B. Variable Definitions

Top3Share $_{i,t}$ is the share of team i 's total payroll allocated to its three highest-paid players during season t , calculated as $\text{Top3Salary}(i,t) / \text{TaxableSalary}(i,t)$. This is the paper's primary measure of payroll concentration. Focusing on the three highest-paid players reflects the fact that NBA payrolls are typically anchored by a small core of maximum or near-maximum contracts, while expressing concentration as a share allows comparisons across seasons despite substantial growth in the salary cap.

Dist_Pct $_{i,t}$ is team i 's taxable salary relative to the luxury tax threshold, calculated as $\text{TaxableSalary}(i,t) / \text{TaxThreshold}(t) - 1$. Positive values indicate teams above the tax line; negative values indicate teams below it.

ApronErat equals 1 for seasons beginning in 2023–24 and 0 otherwise (see footnote 1 regarding the phased-in timing of the apron's full restrictions).

Repeater $_{i,t}$ equals 1 if team i is subject to the NBA's repeater tax during season t , and 0 otherwise.

Win_Pct $_{i,t}$ is team i 's regular-season winning percentage during season t .

Wins_Prior $_{i,t-1}$ is team i 's regular-season winning percentage during the previous season.

Top3_BPM $_{i,t}$ is the minutes-weighted average Box Plus-Minus (BPM) of a team's three highest-paid players.

Top3Yrs_Avg $_{i,t}$ is the salary-weighted average years remaining on a team's three largest contracts, used as a control for contractual lock-in

MidVet_MinShare $_{i,t}$ is the share of total team minutes played by classified mid-level veterans.

N_MidVet $_{i,t}$ is the number of classified mid-level veterans on team i 's roster.

Minutes_per_MidVet $_{i,t}$ is the average minutes played per classified mid-level veteran on team i 's roster.

HighSalVet_MinShare $_{i,t}$ is the share of total team minutes played by classified high-salary veterans.

Taxable Salary $_{i,t}$ is team i 's total taxable payroll during season t .

HighSalVet $_{i,t}$ equals 1 for players with at least four years of NBA service earning more than 13 percent of the league salary cap during season t , and 0 otherwise.

Mid_Veti $_{i,t}$ equals 1 for players with at least four years of NBA service earning between approximately 7 and 13 percent of the league salary cap during season t , and 0 otherwise.²

Rookie_Scale $_{i,t}$ equals 1 for players who were first-round draft selections still within their first four years of NBA service (i.e., still on their rookie-scale contract), and 0 otherwise.

EPM $_{i,t}$ (Estimated Plus-Minus) is a play-by-play-based player impact metric that accounts for teammate quality.³

BPM $_{i,t}$ (Box Plus-Minus) is Basketball-Reference's box-score-based player impact metric.

Teammate_BPM $_{i,t}$ is the leave-one-out, minutes-weighted average BPM of a player's teammates.

C. Summary Stats

Table 1, the first of twelve tables collected in Appendix A, reports summary statistics for the team- and player-level samples. Payroll concentration (Top3Share) averages 52.3 percent, with meaningful variation across teams (SD = 0.102, range: 18.7–79.7 percent) — teams differ substantially in how they allocate payroll. Mid-level veterans account for roughly 13.8 percent of team minutes on average, though this figure is right-skewed (skewness = 0.538) because a number of team-seasons carried no mid-level veterans at all. For those team-seasons, minutes per mid-level veteran is undefined (NA) rather than zero, since there is no player to average over — which is why this measure drops to $N = 384$ rather than the full $N = 450$. At the player level, the sample is weighted toward rookie-scale players (45.3 percent), followed by high-salary veterans (33.0 percent) and mid-level veterans (21.7 percent), reflecting the tiered structure of NBA roster construction under the salary cap.

IV. Empirical Methodology

A. Estimating Equations

² The 7–13 percent range is centered around the value of the NBA's primary non-taxpayer mid-level exception, which averaged approximately 8–9 percent of the salary cap over the sample period, while extending upward to capture similarly experienced complementary veterans earning modestly above the exception. The lower bound excludes minimum-salary and fringe roster players, while the upper bound distinguishes these mid-level veterans from higher-salary veterans.

³ EPM is available only for the 2025–26 season and is used in the paper's closing case study.

The empirical strategy follows the paper's three research questions. I first examine whether teams allocated a greater share of payroll to their highest-paid players following the implementation of the second apron. Conditional on evidence of greater payroll concentration, I then investigate how more concentrated payroll structures affect roster composition, focusing on the use of mid-level veterans. Finally, I evaluate whether these roster-building decisions translate into improved team performance, and whether payroll concentration itself—independent of the quality of the players receiving it—predicts winning. All models include team fixed effects (α_i) and year fixed effects (τ_t), with standard errors clustered at the team level.

Model 1 (Payroll Concentration After the Second Apron):

$$\text{Top3Share}_{i,t} = \beta_0 + \beta_1 \cdot \text{ApronEra}_t + \beta_2 \cdot \text{Wins_Prior}_{i,t} + \beta_3 \cdot \text{Top3Yrs_Avg}_{i,t} + \alpha_i + \varepsilon_{i,t} \quad (1)$$

Model 1 examines whether the introduction of the second apron was associated with changes in how teams allocated payroll across their rosters, specifically whether teams devoted a larger share of total payroll to their three highest-paid players following the policy change. The dependent variable is the share of team payroll allocated to a team's three highest-paid players (Top3Share), while the coefficient on ApronEra captures the average change in payroll concentration after the policy's implementation. Because ApronEra is perfectly collinear with year fixed effects, the model includes team fixed effects but not year fixed effects, and should therefore be interpreted as a descriptive policy comparison rather than the paper's primary identification strategy. To examine whether the distribution of payroll concentration changed even if the league average did not, the model is re-estimated using indicator variables for whether a team's Top3Share exceeds the sample median and, separately, a fixed 55 percent threshold. These alternative specifications test whether highly concentrated payroll structures became more common following the second apron.

Model 2 (Payroll Concentration and Roster Composition):

$$\begin{aligned} \text{MidVet_MinShare}_{i,t} = & \beta_0 + \beta_1 \cdot \text{Top3Share}_{i,t} + \beta_2 \cdot \text{Wins_Prior}_{i,t} + \beta_3 \cdot \text{Top3Yrs_Avg}_{i,t} \\ & + \beta_4 \cdot [\text{Top3Share}_{i,t} \times \text{Era}_t] + \alpha_i + \tau_t + \varepsilon_{i,t} \end{aligned} \quad (2)$$

Model 2 estimates whether greater payroll concentration (Top3Share) is associated with a smaller share of team minutes played by mid-level veterans (MidVet_MinShare). Because major roster decisions occur during the offseason, the primary specifications use a team's prior-season winning percentage (Wins_Prior) and three-year average payroll concentration (Top3Yrs_Avg) to account for existing team quality and contractual persistence. The interaction term (β_4) tests whether this relationship strengthened specifically in the seasons following the second apron's introduction. The model is re-estimated using the number of mid-level veterans (N_MidVet) and average minutes per mid-level veteran (Minutes_per_MidVet) as alternative dependent variables to distinguish whether concentrated payroll structures reduce veteran playing time by carrying fewer mid-level veterans or by reducing the playing time of veterans already on the roster. A final specification adds the share of minutes played by high-salary veterans (HighSalVet_MinShare) as a control to determine whether any reduction in mid-level veteran minutes simply reflects stars occupying a larger share of available minutes or a broader roster-construction effect.

Model 3 (Roster Composition and Team Performance):

$$\text{Win_Pct}_{i,t} = \beta_0 + \beta_1 \cdot \text{MidVet_MinShare}_{i,t} + \beta_2 \cdot \text{Top3Share}_{i,t} + \beta_3 \cdot \text{Top3_BPM}_{i,t} + \alpha_i + \tau_t + \varepsilon_{i,t} \quad (3)$$

Model 3 estimates whether greater reliance on mid-level veterans (MidVet_MinShare) is associated with improved team performance while controlling for payroll concentration (Top3Share) and the quality of a team's three highest-paid players (Top3_BPM). Because payroll concentration and star-player quality are naturally correlated, controlling for Top3_BPM allows the analysis to distinguish the quality of the players receiving concentrated payroll from payroll concentration itself. The model is re-estimated with distance from the luxury tax threshold (Dist_Pct) and, separately, total team payroll (Taxable Salary) to determine whether the relationship between mid-level veteran minutes and winning is simply driven by differences in overall team spending.

Model 4 (Does Concentration Itself Predict Winning?):

$$\text{Win_Pct}_{i,t} = \beta_0 + \beta_1 \cdot \text{Top3Share}_{i,t} + \text{Controls}'_{i,t} \gamma + \alpha_i + \tau_t + \varepsilon_{i,t} \quad (4)$$

Model 4 examines whether payroll concentration itself predicts winning, or whether any relationship between concentrated payroll structures and team performance is entirely attributable to the quality of the players receiving the concentrated payroll. To answer this question, the model is estimated three times using alternative control sets: (i) total team payroll (Taxable Salary), (ii) the quality of the team's three highest-paid players (Top3_BPM), and (iii) both variables simultaneously. Comparing the estimated relationship between payroll concentration and winning across these specifications isolates whether concentrating payroll has any independent association with team performance once differences in overall spending and star-player quality are taken into account.

Model 5 (When Does Concentrating Payroll Around Elite Talent Pay Off?):

$$\text{Win_Pct}_{i,t} = \beta_0 + \beta_1 \cdot \text{Top3Share}_{i,t} + \beta_2 \cdot \text{Top3_BPM}_{i,t} + \beta_3 \cdot [\text{Top3Share}_{i,t} \times \text{Top3_BPM}_{i,t}] + \beta_4 \cdot \text{Taxable_Salary}_{i,t} + \alpha_i + \tau_t + \varepsilon_{i,t} \quad (5)$$

Model 5 tests whether payroll concentration produces diminishing marginal returns to talent quality — that is, whether an additional unit of star-player quality becomes less valuable as a larger share of payroll is already committed to a team's highest-paid players. The model relaxes Model 4's assumption that quality's effect on winning is constant across teams by introducing an interaction between Top3Share and Top3_BPM. The coefficient of interest is β_3 : a negative and significant interaction indicates that the marginal benefit of improving talent quality shrinks as payroll becomes more concentrated, and — read the other way — that concentrating payroll further becomes less beneficial, or potentially costly, as a team's existing talent quality rises. The model is then re-estimated with MidVet_MinShare added as a further control, testing whether this interaction operates in part through the same mid-level-veteran crowd-out mechanism documented in Model 2.

B. Identifying Assumptions and Threats to Identification

First, the analysis assumes conditional exogeneity. After accounting for team fixed effects, year fixed effects, and the included control variables, the remaining variation in the explanatory variables is assumed to be unrelated to unobserved determinants of the outcome variables. Under this assumption, the estimated coefficients can be interpreted as the partial relationships between payroll allocation, roster construction, and team performance.

Second, the analysis assumes that roster construction decisions precede regular-season performance. Payroll allocation and roster composition are largely determined during the offseason, while team performance is realized over the subsequent regular season. This timing supports interpreting payroll allocation and roster construction as preceding team performance rather than responding to it. The inclusion of prior-season winning percentage further accounts for pre-existing differences in team quality that may influence subsequent roster decisions.

Third, the analysis assumes that the quality of a team's highest-paid players is adequately captured by the minutes-weighted average Box Plus-Minus (Top3_BPM). Because elite players both command larger salaries and contribute disproportionately to team success, payroll concentration and player quality are naturally correlated. Controlling for Top3_BPM allows the analysis to distinguish differences in payroll concentration from differences in the quality of the concentrated talent, while Model 5 further assumes that the interaction between Top3Share and Top3_BPM captures how player quality moderates the relationship between payroll concentration and team performance.

V. Payroll Concentration and Roster Construction

A. The Second Apron and Payroll Concentration

The first empirical test examines whether teams allocated a greater share of payroll to their highest-paid players following the introduction of the second apron. Table 2 (Appendix A) reports estimates from both Model 1 specifications — the league-average level of concentration and the share of teams exceeding a given concentration threshold. Figures 1 and 2 (Appendix B) illustrate these two patterns directly: Figure 1 shows the league-average level of payroll concentration over time, and Figure 2 shows the share of teams allocating more than half of their payroll to their three highest-paid players. Although average payroll concentration increased modestly after the second apron's implementation, the estimated increase is not statistically significant after controlling for persistent differences across teams ($\beta = 0.016$, $p = 0.13$; Table 2, Panel A, Column 1). At the league level, therefore, the second apron was not associated with a meaningful increase in average payroll concentration.

Despite this limited change in the league-wide average, many more teams adopted highly concentrated payroll structures following the second apron. As shown in Figure 2, the share of team-seasons allocating more than

half of payroll to their three highest-paid players increased from 60.0 percent before the policy to 72.2 percent afterward ($p = 0.039$). This pattern holds using the sample median as an alternative cutoff (52.6 percent): the share of team-seasons above the median rose from 47.5 percent to 60.0 percent ($p = 0.040$), while the share exceeding a fixed 55 percent threshold rose from approximately 39 percent to approximately 52 percent ($p = 0.049$) (Table 2, Panel B). Similar increases appear across a broad range of concentration thresholds, including cutoffs above 60 percent; statistical significance becomes harder to detect at the highest thresholds given the small number of observations in the short post-apron period, though the estimated effects remain consistently positive. The second apron, in other words, did not substantially raise average payroll concentration across the league, but it did coincide with a broader shift toward more teams operating with highly concentrated payrolls.

Both figures also reveal a pronounced temporary spike in payroll concentration during the 2019 offseason, raising the possibility that the post-2023 pattern simply reflects a pre-existing trend or another short-lived wave of superstar movement rather than a structural change. Excluding the three teams with the most concentrated payrolls in 2018-19 removes only 0.0141 of the roughly 0.077 increase that year (Table 2, Panel C), indicating the spike was not driven by a handful of outliers — it was broadly shared across the league, consistent with moves such as Kevin Durant and Kyrie Irving joining the Brooklyn Nets and Kawhi Leonard and Paul George joining the Los Angeles Clippers (CBS Sports, 2019). The spike itself receded within two seasons, but concentration did not fully return to its pre-2019 level: the three seasons immediately preceding the apron (2020-2022) settled modestly above the earlier baseline. This could reflect a genuine, broader drift toward concentration that predates the second apron, or simply a handful of the same 2019 rosters — most notably the Clippers' Leonard-George pairing — persisting a few additional seasons rather than a league-wide shift; the data here cannot fully distinguish between these explanations. What the data do show clearly is that the post-2023 increase has persisted for the entire available post-apron sample without receding the way the 2019 spike did, distinguishing it from a purely temporary fluctuation, even if it is not fully separable from a modest upward drift already underway before the apron's introduction.

B. Payroll Concentration and Roster Composition

Having established that more teams adopted concentrated payroll structures following the second apron, I next examine how payroll concentration affects roster composition more broadly. Specifically, I investigate whether

teams with more concentrated payrolls devote fewer minutes to mid-level veterans — experienced rotation players likely to be displaced as payroll becomes increasingly concentrated.

Estimating Model 2, the results reveal a significant negative relationship between payroll concentration and the share of team minutes played by mid-level veterans. A 10-percentage-point increase in Top3Share is associated with approximately a 2.8-percentage-point reduction in mid-level veteran minute share ($\beta = -0.283$, $p = 0.0045$; Table 3, Column 1). Over an 82-game season, this corresponds to roughly 557 fewer team minutes — almost twelve full games of playing time. This relationship is illustrated in Figure 3, Panel A (Appendix B).

One remaining concern is that two teams could share the same level of payroll concentration while differing substantially in total spending — a large-market team willing to spend deep into the tax could match another team's concentration percentage while still having far more absolute dollars left over for the rest of the roster after paying its stars. If that were driving the result, the relationship between concentration and mid-level veteran minutes would really be a relationship between total spending and mid-level veteran minutes in disguise. To rule this out, I re-estimate the model controlling separately for luxury tax position and total taxable payroll — directly holding total spending fixed so that any remaining relationship must come from concentration itself, not from how much money a team has overall. The coefficient moves only slightly, to -0.316 controlling for tax-threshold distance and -0.318 controlling for raw taxable salary (Table 3, Columns 2–3): even comparing teams with the same overall payroll, more concentrated teams still devote less to mid-level veterans.

This finding holds across the full sample, not only in the years since the second apron. It is broadly consistent with a simple budget-constraint logic: money committed to a team's highest-paid players is money unavailable for the rest of the roster. The second apron's specific restrictions on the taxpayer mid-level exception and salary aggregation in trades — the primary tools teams have historically used to sign and retain mid-level veterans — plausibly sharpen this tradeoff further for the most concentrated teams. Section V.D tests this directly by examining whether the relationship between payroll concentration and mid-level veteran roster presence grew stronger specifically in the seasons following the second apron's introduction. The underlying relationship documented here, however, predates the second apron and holds across the full sample regardless of that result.

C. Is the Decline in Mid-Level Veterans Mechanical?

One possible explanation for the relationship documented above is that it is purely mechanical. Because star players generally play more minutes, allocating a larger share of payroll to elite talent may simply leave fewer minutes available for everyone else.

Re-estimating Model 2 with `HighSalVet_MinShare` added as a control, the estimated effect of payroll concentration falls modestly, from -0.283 to -0.212 , and remains statistically significant ($p = 0.023$; Table 3, Columns 1 and 4). This suggests that roughly a quarter of the decline in mid-level veteran minutes reflects stars mechanically occupying a larger share of available playing time; the majority of the relationship — roughly three-quarters of the original effect — persists even after accounting for star minutes directly, suggesting that payroll concentration reflects broader roster-construction decisions rather than merely a mechanical redistribution of minutes within the existing roster. The section that follows identifies exactly where that roster-construction effect shows up: whether teams carry fewer mid-level veterans outright, or simply play the ones they have fewer minutes each.

D. Fewer Mid-Level Veterans or Less Playing Time?

The previous results establish that concentrated payroll structures reallocate minutes away from mid-level veterans. An important remaining question is what actually drives this decline in minutes: do teams retain the same mid-level veterans but reduce their playing time, or do they simply carry fewer such players on their rosters?

The evidence strongly supports the latter explanation. Re-estimating Model 2 using the number of rostered mid-level veterans (`N_MidVet`) and the average minutes played per mid-level veteran (`Minutes_per_MidVet`) as alternative outcomes, teams with more concentrated payrolls roster significantly fewer mid-level veterans: a 10-percentage-point increase in payroll concentration (`Top3Share`) predicts roughly 0.44 fewer mid-level veterans rostered ($\beta = -4.44$, $p < 0.001$; Table 4, Column 1). Because a typical roster in the sample carries only two to three mid-level veterans at any given time, losing nearly half a player per standard-deviation shift in concentration represents a meaningful share of a team's entire mid-level veteran presence — the equivalent of one in every two or three concentrated teams effectively losing an entire rotation spot at that tier. Payroll concentration has no statistically significant relationship with the average minutes played by each remaining mid-level veteran ($\beta = 0.025$, $p = 0.28$; Table 4, Column 2). This roster-headcount relationship is illustrated in Figure 3, Panel B (Appendix B).

This distinction matters because it identifies the actual mechanism behind the roster changes documented above. Rather than shortening the rotation of the veterans they keep, teams appear to reduce roster depth itself by carrying fewer of them in the first place. This finding holds across the full sample, consistent with the same budget-constraint logic discussed earlier: as payroll becomes more concentrated, a smaller share of roster spots and rotation minutes go to mid-level veterans. Adding an interaction between payroll concentration and the apron-era indicator — testing whether this relationship strengthened after the second apron — returns a result directionally consistent with a stronger effect (roughly half as large as the baseline coefficient) but not statistically significant ($p = 0.20$; Table 4, Column 3). This is unsurprising given only three apron-era seasons to estimate from, and is consistent with the second apron intensifying an existing relationship rather than creating a new one, by further restricting the roster-building tools teams use to retain this tier of player.

Given that more teams now operate with concentrated payrolls, and that such concentration is associated with carrying fewer mid-level veterans, the following section examines how this roster structure — concentration at the top, with reduced reliance on mid-level veterans — influences team performance.

VI. Roster Construction and Team Performance

A. How Do Mid-Level Veterans Impact Winning?

The previous section demonstrated that greater payroll concentration is associated with fewer minutes played by experienced mid-level veterans and, more importantly, with carrying fewer such players on the roster. The next question is how these changes in roster composition factor into team performance.

Estimating Model 3, teams allocating a larger share of minutes to mid-level veterans consistently win more games across all specifications. A 10-percentage-point increase in mid-level veteran minute share is associated with approximately a 2.8-percentage-point increase in winning percentage in the baseline specification, which already controls for payroll concentration and the quality of a team's three highest-paid players ($\beta = 0.277$, $p < 0.001$; Table 5, Column 1). This relationship is illustrated in Figure 4 (Appendix B). Importantly, this relationship remains statistically significant after also controlling for total team payroll: the coefficient changes very little whether that control is luxury tax position ($\beta = 0.264$) or total taxable payroll ($\beta = 0.265$), indicating that the result is not simply driven by higher-spending teams rostering more mid-level veterans (Table 5, Columns 2–3).

These findings suggest that experienced complementary veterans contribute to team success independently of overall payroll and star quality. While star players remain essential, the results suggest that maintaining productive roster depth provides measurable value beyond the contributions of a team's highest-paid players alone.

B. Does Payroll Concentration Itself Improve Team Performance?

Section II showed that prior studies generally find a positive relationship between payroll or talent concentration and winning, at least up to a point. At first glance, the results here appear to reach a similar conclusion. When team performance is regressed on payroll concentration while controlling only for total payroll, more concentrated payroll structures are associated with higher winning percentages ($\beta = 0.350$, $p < 0.001$; Table 6, Column 1).

That relationship disappears once the quality of the players receiving the concentrated payroll is taken into account. Controlling for the quality of a team's three highest-paid players reduces the estimated effect of payroll concentration from $\beta = 0.350$ to $\beta = 0.041$ (Table 6, Column 2), a coefficient statistically indistinguishable from zero. Adding total payroll as an additional control leaves the estimate essentially unchanged ($\beta = -0.009$, $p = 0.914$; Table 6, Column 3).

This progression illustrates a straightforward case of omitted variable bias. Elite players both command larger salaries and contribute disproportionately to winning, making payroll concentration and player quality naturally correlated. Once player quality is held constant, payroll concentration itself has essentially no average relationship with team performance.

C. When Does Concentrating Payroll Around Elite Talent Pay Off?

The previous section showed that payroll concentration, by itself, has no independent relationship with winning once player quality is taken into account. The remaining question is whether the value of elite talent depends on how much of a team's payroll is committed to that talent.

To examine this possibility, I estimate an interaction between the quality of a team's three highest-paid players (Top3_BPM) and payroll concentration (Top3Share) while controlling for total payroll. The interaction is negative and statistically significant ($\beta = -0.059$, $p = 0.014$; Table 7, Column 1), indicating that the relationship between star-player quality and winning becomes weaker as payroll becomes more concentrated.

The practical implications follow directly. For a team allocating roughly 40 percent of its payroll to its three highest-paid players, a one-point increase in that group's quality — whether through player development, a like-for-like swap of similarly paid players, or ordinary year-to-year performance variation — is associated with approximately a 5.3 percentage-point increase in winning percentage, about 4.4 additional wins over an 82-game season. For an otherwise identical team allocating 60 percent of its payroll to an equally improved top three, that same quality gain is associated with only a 4.1 percentage-point increase, or roughly 3.4 additional wins (Table 7, Marginal Effects). Talent quality always helps team performance, but its marginal benefit is smaller for teams that have already committed a larger share of the cap to a small core.

This is not because elite players are individually less valuable at higher concentration. One plausible explanation is that concentrating a larger share of a fixed payroll around the same core leaves fewer resources for the rest of the roster. A robustness check testing this directly adds the share of minutes played by mid-level veterans (MidVet_MinShare) to the model. Once this measure of roster depth is included, the interaction coefficient falls by roughly one-third (from $\beta = -0.059$ to $\beta = -0.039$) and becomes only marginally significant ($p = 0.081$; Table 7, Column 2) — evidence that a meaningful part of this pattern operates through the mechanism identified earlier in the paper, where more concentrated payroll structures reduce reliance on mid-level veterans, but not evidence that this fully explains it.

Taken together, these findings suggest that payroll concentration does not diminish the value of elite players themselves — a team with more top-end talent is expected to win more, regardless of how its payroll is split. Instead, they point to a genuine tradeoff: every dollar committed to an already-expensive core is a dollar unavailable to build the roster depth around it, and as concentration rises, that tradeoff becomes steeper. This does not mean concentrating payroll around elite talent is a mistake — it means front offices must weigh it deliberately, case by case, against the specific value of the alternative: filling out the roster with quality depth instead. Where that balance falls depends on context a single coefficient cannot capture — how the rest of the roster is already built, how much cap flexibility remains, and how close a team is to a competitive window worth pushing for. The Boston Celtics' decision on Jaylen Brown, examined next, illustrates exactly this kind of judgment call in practice.

D. Summary of Empirical Findings

Taken together, the results suggest a clear sequence linking the second apron, roster construction, and team performance. Following the introduction of the second apron, substantially more teams adopted highly concentrated payroll structures, even though league-average payroll concentration changed little. Teams with more concentrated payrolls consistently rostered fewer experienced mid-level veterans ($\beta = -4.44$, $p < 0.001$), while payroll concentration had no significant relationship with the playing time of the veterans who remained ($\beta = 0.025$, $p = 0.28$). Greater reliance on mid-level veterans, in turn, was associated with improved team performance even after accounting for payroll concentration, total payroll, and the quality of a team's highest-paid players ($\beta = 0.277$, $p < 0.001$).

Perhaps most importantly, payroll concentration itself has no independent average relationship with winning once player quality is taken into account. The apparent positive relationship between concentration and team success ($\beta = 0.350$, $p < 0.001$) disappears after controlling for the quality of a team's three highest-paid players ($\beta = -0.009$, $p = 0.914$), indicating that previous estimates largely reflected omitted variable bias rather than an independent benefit of concentrating payroll.

Moreover, the results further show that the value of talent quality is itself contingent on how concentrated a team's payroll already is. Improved talent quality is consistently associated with better team performance, but those gains grow smaller as a larger share of payroll is already committed to a team's top three players — and a robustness check indicates part of this pattern is explained by teams rostering fewer mid-level veterans, not simply by any weakening in the value of quality itself.

Collectively, these findings suggest that payroll concentration does not diminish the value of elite players themselves — a team with better talent is still expected to win more, regardless of how its payroll is split. Instead, they point to a genuine tradeoff: every dollar committed to an already-expensive core is a dollar unavailable to build the roster depth around it, and as concentration rises, that tradeoff becomes steeper rather than staying constant. This does not mean concentrating payroll around elite talent is a mistake — it means front offices must weigh it deliberately, case by case, against the specific value of the alternative: filling out the roster with quality depth instead. Where that balance falls depends on context a single coefficient cannot capture — how the rest of the roster is already built, how much cap flexibility remains, and how close a team is to a competitive window worth pushing for. The Boston Celtics' decision on Jaylen Brown, examined next, illustrates exactly this kind of judgment call in practice.

VII. Case Study: Jaylen Brown and the Boston Celtics

While the previous sections identify league-wide relationships between payroll concentration, roster construction, and team performance, NBA front offices ultimately face these tradeoffs through individual personnel decisions. The Boston Celtics provide a useful case study because they entered the second apron with one of the league's most talented — and most expensive — rosters. Jaylen Brown's five-year supermax extension, signed in July 2023 after earning All-NBA honors, made him the highest-paid player in NBA history at the time (NBA.com, 2023) and substantially increased Boston's long-term payroll commitments. In the summer of 2026, Boston traded Brown, receiving Paul George and future draft compensation in return (NBA.com, 2026) — a decision that gives Boston greater financial flexibility in the near future, once George's contract comes off the books.

The purpose of this case study is not simply to evaluate Brown as a player, or to judge Boston's trade in hindsight. Rather, it asks whether Brown's on-court production justified the share of payroll he occupied under the second apron, and whether the evidence presented here helps explain why Boston made the decision it did. The league-wide analysis above suggests that payroll concentration itself is neither inherently beneficial nor harmful — success depends on whether the players receiving the concentrated investment generate sufficient value to justify the resulting reduction in roster depth. The Boston Celtics provide a useful illustration of this tradeoff.

I begin by examining Estimated Plus-Minus (EPM), a player evaluation metric that estimates overall impact using lineup and play-by-play data rather than traditional box-score statistics alone. Unlike many player metrics, EPM already adjusts for the quality of a player's teammates, making it particularly useful for comparing players across different roster contexts. During the 2025-26 season, Brown posted an overall EPM of +2.2, which ranked in the 89th percentile league-wide, while his Offensive EPM of +2.3 comfortably exceeded league average. Viewed in isolation, these results suggest Brown remained one of the NBA's better players.

The more economically relevant question, however, is whether that level of production matched the offensive role and financial resources committed to him. During the 2025-26 season, Brown posted a 35.2% usage rate, the third-highest in the NBA, placing him alongside players such as Nikola Jokić, Luka Dončić, Giannis Antetokounmpo, and Shai Gilgeous-Alexander. Players entrusted with that level of offensive responsibility are generally expected to produce elite impact.

To evaluate whether Brown met those expectations, I regress both EPM and Offensive EPM on usage rate and salary across the classified high-salary veteran population introduced earlier in the paper (N = 248 player-seasons pooled across the comparison population; Table 8) — veteran players with at least four prior years of NBA service earning more than 13% of the salary cap during a given season. Each measure is estimated two ways: a linear specification, and a quadratic specification allowing returns to usage to accelerate at the very top of the distribution, where a small number of true superstars carry disproportionate offensive loads. Comparing each player's actual production with each model's predicted production yields a residual measuring how far above or below expectation that player performed relative to comparable high-salary veterans. Positive residuals indicate players who generated more value than expected given their offensive role, while negative residuals indicate players who produced less than expected.

Across all four specifications, Brown ranked 2nd-to-9th (of 84) by residual size, falling roughly 2.3 points below expectation under the linear models and 3.2–3.8 points under the quadratic models (Table 9). Put differently, relative to other high-salary veterans carrying similar offensive roles and financial commitments, Brown was statistically one of the league's most underwhelming performers — an All-NBA-caliber player whose statistical production nonetheless fell well short of what teams typically receive from someone occupying such a large share of both the offense and the payroll. This comparison is illustrated in Figure 5 (Appendix B), which plots usage rate against EPM residual from the quadratic specification for the full population of classified high-salary veterans; an interactive version of this figure, allowing readers to hover over individual players, is linked alongside it in Appendix B.

One potential concern is that Brown's 2025-26 season may not reflect his typical role. Jayson Tatum missed much of the season while recovering from an Achilles injury, resulting in Brown's highest usage rate of his career. To determine whether these findings reflect one unusual season or a more persistent pattern, I repeat the analysis for each of the previous three seasons, when Tatum remained healthy and Brown occupied his more typical offensive role. Because freely available EPM data were only available for the 2025-26 season, these earlier analyses use Box Plus-Minus (BPM) instead. Unlike EPM, BPM does not inherently adjust for surrounding talent, so each regression includes a leave-one-out, minutes-weighted measure of teammate quality (Teammate_BPM) to isolate Brown's individual performance from the quality of the roster around him (Tables 10–11, linear and quadratic specifications, respectively).

The results reveal a remarkably consistent pattern (Table 12, Panel A). In 2022-23, Brown's BPM fell 2.34 points below expectation, ranking 13th-worst among 84 classified high-salary veterans. In 2023-24, the shortfall widened to 2.70 points, ranking 18th-worst among 87. By 2024-25, Brown's BPM fell 4.80 points below expectation, the second-largest negative residual among 93 high-salary veterans league-wide. Over the same period, Brown's raw BPM steadily declined from 2.5 in 2020-21 to 0.0 in 2024-25, even as his salary more than doubled, from \$23.7 million to \$49.2 million (Table 12, Panel B). Together, these results suggest that Brown's 2025-26 performance was not an isolated outlier, but the continuation of a broader trend in which his statistical production increasingly lagged behind what would be expected from a player with his salary and offensive role.

These findings come with an important caveat. EPM and BPM don't fully capture certain contributions to winning — things like defensive gravity, secondary playmaking, and floor spacing are only partially reflected in these metrics, while qualities like leadership go largely unmeasured. Brown's statistical shortfall relative to his salary and role doesn't mean his overall value to the team was similarly limited; he was, after all, Finals MVP of Boston's 2024 championship run. The claim here is narrower: using the same measurable standard applied to every player in this paper, Brown's production fell short of what his pay and usage predicted — not a full verdict on his value to the team.

Derrick White provides an informative comparison because nearly every external factor remained constant. Both players competed within the same organization, under the same coaching staff, and alongside the same teammates. During the 2025-26 season, White earned roughly half of Brown's salary (\$28.1 million versus \$53.1 million) while carrying a substantially smaller offensive role (21.8% usage versus 35.2%). In the same usage-and-salary-adjusted framework, White's residual was positive across every specification — for example, +3.39 in the linear EPM model, compared to Brown's -2.34 (Table 9). Relative to what his usage rate and salary would predict, White outperformed expectation while Brown fell short of it — at roughly half the cost. This gap is far more costly under the second apron than it would have been under previous rules.

Viewed through this paper's broader framework, Brown's contract illustrates the broader economic tradeoff created by the second apron rather than serving as a judgment of Brown's ability. Brown remained an All-NBA caliber player and an important contributor to a championship team. The relevant question is not whether he deserved to be highly paid, but whether committing a supermax-level share of payroll to a player producing below

expectation relative to his salary and offensive role represented the organization's most efficient use of increasingly scarce financial resources.

Taken together, the Boston case study reinforces the central conclusion of this paper. Concentrating payroll is not inherently a successful or unsuccessful roster-building strategy. Rather, concentrating payroll creates a tradeoff between investing in elite talent and maintaining roster depth. When the players receiving the largest share of payroll generate truly elite value, that tradeoff may be worthwhile. When they do not, teams sacrifice the roster depth that this paper finds is independently associated with winning.

VIII. Conclusion

The introduction of the NBA's second apron has changed the economics of roster construction. Rather than changing the importance of star players themselves, the new Collective Bargaining Agreement has changed the cost of building around them. Concentrating payroll increasingly comes with an opportunity cost: committing more resources to a small group of highly paid players leaves fewer resources for experienced depth elsewhere on the roster.

The league-wide average level of payroll concentration did not increase significantly after the second apron's introduction, but a larger share of teams adopted highly concentrated structures — a broader shift toward top-heavy rosters, even without a uniform league-wide push.

The consequences show up in roster composition, not playing time. A 10-percentage-point increase in payroll concentration is associated with roughly 0.44 fewer mid-level veterans, with essentially no effect on the minutes played by those who remain. Meanwhile, teams that devote more minutes to mid-level veterans win more games, even controlling for total payroll and the quality of their top three players — evidence that depth is an economically meaningful driver of performance, and that concentration at the top carries a real cost elsewhere.

Concentration itself does not independently predict winning. A positive relationship between concentration and winning appears when only total payroll is controlled for, but disappears once the quality of the players receiving that payroll is accounted for. Its value depends not simply on whether the concentrated players are good, but on when (and whether) concentrating around them actually pays off, given what teams give up elsewhere to do it.

The interaction results sharpen this tradeoff: higher-quality top-three players consistently improve performance, but the marginal gain shrinks as more payroll is already committed to them. A one-BPM improvement in top-three quality raises winning percentage by roughly 5.3 percentage points when the top three account for 40 percent of payroll, versus 4.1 percentage points at 60 percent. A robustness check ties this pattern to reduced reliance on mid-level veterans, reinforcing that concentrating further into an already expensive core carries rising opportunity costs.

The implication isn't that teams should avoid paying elite players — it's that the value of further investment in elite talent depends on the roster around it. Once a team has committed heavily to a small core, additional concentration leaves less for complementary players whose contributions independently drive winning. Under the second apron, this tradeoff matters more, since restrictions on roster-building tools make lost depth harder to replace.

The Jaylen Brown case study illustrates this tradeoff at the individual-team level. Brown remained an excellent player and an important contributor to Boston's championship team, but his statistical production relative to his salary and offensive role fell short of what comparable high-salary players typically produced. In an environment where Boston had fewer mechanisms available to replenish its supporting cast, the opportunity cost of that commitment became substantially larger. The case therefore illustrates the broader lesson of the paper: the relevant question is not simply whether a highly paid player is good, but whether his production is sufficient to justify the roster resources that must be sacrificed to pay him.

This study has several limitations. Most importantly, the second apron has been in effect for only three seasons, leaving relatively few post-policy observations from which to estimate its long-run effects. The available evidence is nevertheless consistent across multiple specifications and robustness checks, but additional seasons will provide a stronger basis for distinguishing a persistent structural change from a short-run adjustment. Future research can build on these findings by examining whether the shift toward concentrated payroll structures and reduced reliance on mid-level veterans persists as teams continue adapting to the new CBA. Future work could also examine the alternative roster-building strategies teams use to replace the depth sacrificed by concentrated payrolls, including greater reliance on rookie-scale players, internal player development, and draft capital. Finally, as more seasons of data become available, researchers could examine whether these adjustments ultimately affect long-run team performance, rather than only the roster composition and within-season relationships identified in this study.

Ultimately, the second apron has not made star talent less important. It has made the cost of concentrating resources around that talent more consequential. As financial flexibility becomes increasingly constrained, successful roster construction depends not simply on acquiring elite players, but on balancing the value of those players against the depth that must be sacrificed to support them. The second apron's most important effect may not be how much teams spend, but what they are forced to sacrifice when they decide where that spending goes.

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Appendix A:

Table 1: Summary Statistics

Panel A: Team-Level Variables ($N = 450$ Team-Seasons)

Variable	N	Mean	SD	Min	Median	Max	Skewness
Payroll Concentration (Top3Share)	450	0.523	0.102	0.187	0.526	0.797	-0.273
Tax Distance (Dist_Pct)	450	-0.042	0.119	-0.400	-0.020	0.430	0.298
Win Percentage	450	0.500	0.151	0.106	0.514	0.890	-0.201
Top-3 Player Quality (Top3_BPM)	450	1.374	2.101	-5.267	1.283	6.667	0.000
Mid-Level Veteran Minute Share	450	0.138	0.104	0.000	0.124	0.484	0.538
Mid-Level Veterans Rostered	450	1.758	1.213	0.000	2.000	6.000	0.537
Minutes per Mid-Level Veteran	384	0.078	0.028	0.002	0.079	0.146	-0.263
High-Salary Veteran Minute Share	450	0.267	0.131	0.000	0.264	0.665	0.098
Taxable Salary (\$M)	450	116.11	41.32	48.05	117.33	215.26	0.308
3-Year Avg. Concentration	450	1.522	0.567	0.382	1.461	3.696	0.511
Prior-Season Win %	450	0.500	0.150	0.106	0.512	0.890	-0.193

Panel B: Player-Level Variables ($N = 3,645$ Player-Seasons)

Variable	N	Mean	SD	Min	Median	Max	Skewness
Age	3,645	25.67	4.30	19.00	25.00	41.00	0.570
Minutes per Game (MPG)	3,645	25.18	8.10	2.33	26.72	39.40	-0.600
Share of Team Minutes	3,645	0.081	0.039	0.001	0.087	0.161	-0.298
Salary (\$M)	3,645	10.70	10.66	0.85	7.00	59.61	1.546
Box Plus-Minus (BPM)	3,645	-0.19	3.21	-18.20	-0.30	14.20	0.203
Usage Rate (%)	3,645	20.40	5.52	3.70	19.80	41.70	0.440

*Panel C: Player-Tier Composition
(Player-Level, $N = 3,645$)*

Tier	N	% of Sample
Mid-Level Veterans	791	21.7%
High-Salary Veterans	1,203	33.0%
Rookie-Scale Players	1,651	45.3%

Notes: Team-level sample spans all 30 NBA franchises over 15 seasons (2011-12 through 2025-26), $N = 450$ team-seasons. Player-level sample spans the same period, $N = 3,645$ classified player-seasons (mid-level veteran, high-salary veteran, or rookie-scale). Taxable Salary and Salary are reported in \$ millions for readability; regression coefficients in subsequent tables use raw dollar units unless noted. Minutes per Mid-Level Veteran is undefined (missing) for team-seasons rostering zero mid-level veterans, hence $N = 384$.

Table 2: Payroll Concentration After the Second Apron*Panel A: Regression Estimates*

Dependent Variable:	(1) Top3Share Mean Level	(2) Median_Cutoff Median Threshold	(3) Fixed55_Cutoff 55% Threshold	(4) Above_Half 50% Threshold
Era	0.0163 (0.0105)	0.1245** (0.0579)	0.1272** (0.0620)	0.1218** (0.0563)
Wins_Prior	0.1591*** (0.0316)	1.189*** (0.1568)	1.211*** (0.1591)	1.026*** (0.1647)
Top3Yrs_Avg	0.0880*** (0.0090)			
Team FE	Yes	Yes	Yes	Yes
Observations	450	450	450	450
R^2	0.502	0.257	0.255	0.213
Within R^2	0.404	0.129	0.136	0.103

Panel B: Threshold-Robustness Sweep

Cutoff	Cutoff Value	Pre-Apron %	Apron-Era %	Coef. (Era)	SE	p -value
25th pctile (placebo)	0.464	73.3	81.1	0.077	0.041	0.067
0.50	0.500	60.0	72.2	0.122	0.056	0.039
Median	0.526	47.5	60.0	0.124	0.058	0.040
0.55	0.550	39.4	52.2	0.127	0.062	0.049
0.60	0.600	20.6	27.8	0.072	0.063	0.263
0.65	0.650	8.1	12.2	0.042	0.049	0.399
75th pctile	0.591	23.3	32.2	0.089	0.058	0.135

Panel C: 2019 Concentration Spike — Top-3 Team Exclusion

	Mean Top3Share
League average, all 30 teams (2018-19)	0.5034
League average excluding top 3 teams (Rockets, Celtics, Thunder)	0.4892
Difference attributable to top 3 teams	0.0141

Notes: Team fixed effects only; Era is perfectly collinear with year fixed effects, so Panel A is a descriptive policy comparison rather than the paper's primary identification strategy. Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Panel B reports the share of team-seasons above each Top3Share cutoff, pre- vs. post-apron, with the associated linear-probability coefficient on Era (team FE, clustered by team). The 25th-percentile row is a placebo. Panel C decomposes the 2018-19 year-over-year spike in league-average Top3Share; excluding the three most concentrated teams that season removes only 0.0141 of the roughly 0.077 increase, indicating the spike was broadly shared rather than driven by a handful of outliers.

Table 3: Payroll Concentration and Mid-Level Veteran Minute Share

Dependent Variable:	MidVet_MinShare			
	(1) Base	(2) + Dist_Pct	(3) + Taxable Salary	(4) + HighSalVet Minutes
Top3Share	−0.2834*** (0.0921)	−0.3155*** (0.0971)	−0.3177*** (0.0977)	−0.2119** (0.0879)
Wins_Prior	0.2128*** (0.0276)	0.1882*** (0.0332)	0.1850*** (0.0292)	0.2444*** (0.0336)
Top3Yrs_Avg	−0.0049 (0.0125)	−0.0042 (0.0126)	−0.0046 (0.0125)	−0.0025 (0.0121)
Top3Share × Era	−0.1313 (0.1203)	−0.1283 (0.1201)	−0.1504 (0.1212)	−0.1303 (0.1208)
Dist_Pct		0.0907 (0.0676)		
Taxable Salary			$9.45 \times 10^{-10} *$ (5.1×10^{-10})	
HighSalVet_MinShare				−0.1363** (0.0506)
Team FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	450	450	450	450
R^2	0.218	0.223	0.226	0.234
Within R^2	0.102	0.107	0.111	0.120

Notes: Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column (4) tests whether the minute-share decline is mechanical (stars occupying more minutes) rather than a broader roster-construction choice: Top3Share's coefficient falls from −0.283 to −0.212 but remains significant, indicating roughly a quarter of the effect is mechanical and the remainder reflects a real roster-composition response.

Table 4: Fewer Mid-Level Veterans, or Less Playing Time?

Dependent Variable:	(1)	(2)	(3)
	N.MidVet Headcount	Minutes_per_MidVet Minutes per Player	N.MidVet Headcount $\times$ Era
Top3Share	-4.453*** (0.8084)	0.0254 (0.0221)	-4.082*** (0.9540)
Wins.Prior	2.083*** (0.3156)	0.0373*** (0.0094)	2.063*** (0.3002)
Top3Yrs.Avg	-0.1059 (0.1381)	0.0035 (0.0027)	-0.0961 (0.1342)
Top3Share $\times$ Era			-2.050 (1.525)
Team FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	450	384	450
R^2	0.246	0.215	0.250
Within R^2	0.125	0.083	0.129

Notes: Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column (2) drops 66 team-seasons rostering zero mid-level veterans, for whom minutes-per-player is undefined. A 10-percentage-point increase in Top3Share predicts 0.44 fewer rostered mid-level veterans ($p < 0.001$) with no significant change in minutes played by those retained ($p = 0.28$), indicating the decline in mid-level veteran minutes operates through roster headcount rather than reduced playing time per player. Column (3)'s Era interaction is directionally consistent with a stronger post-apron effect but is not statistically significant ($p = 0.20$).

Table 5: Mid-Level Veteran Minutes and Team Performance

Dependent Variable:	Win.Pct		
	(1) Baseline	(2) + Tax Position	(3) + Total Payroll
MidVet_MinShare	0.2773*** (0.0622)	0.2644*** (0.0604)	0.2652*** (0.0610)
Top3Share	0.1429 (0.0871)	0.0911 (0.0839)	0.0985 (0.0832)
Top3_BPM	0.0425*** (0.0034)	0.0403*** (0.0037)	0.0407*** (0.0037)
Dist.Pct		0.1456** (0.0572)	
Taxable Salary			1.05×10^{-9} ** (4.81×10^{-10})
Team FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	450	450	450
R^2	0.566	0.572	0.571
Within R^2	0.458	0.466	0.464

Notes: Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. A 10-percentage-point increase in mid-level veteran minute share is associated with a 2.8-percentage-point increase in win percentage in the baseline specification, robust to controlling for luxury-tax position and total team payroll.

Table 6: Payroll Concentration and Team Performance

Dependent Variable:	Win_Pct		
	(1) Total Payroll Only	(2) Star Quality Only	(3) Both
Top3Share	0.3503*** (0.0772)	0.0409 (0.0833)	-0.0086 (0.0794)
Taxable Salary	3.18×10^{-9} *** (6.06×10^{-10})		1.31×10^{-9} *** (4.48×10^{-10})
Top3_BPM		0.0473*** (0.0034)	0.0448*** (0.0038)
Team FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	450	450	450
R^2	0.350	0.537	0.545
Within R^2	0.188	0.422	0.432

Notes: Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column (1) shows the apparent positive relationship between concentration and winning when only total payroll is controlled for; this relationship falls to a coefficient statistically indistinguishable from zero once star-player quality (Top3_BPM) is controlled for in Columns (2)-(3), consistent with omitted-variable bias rather than an independent benefit of payroll concentration.

Table 7: Does Concentration's Effect on Winning Depend on Talent Quality?

Dependent Variable:	Win_Pct	
	(1) Baseline Interaction	(2) + Mid-Vet Minute Share
Top3Share	0.0306 (0.0819)	0.1187 (0.0831)
Top3_BPM	0.0768*** (0.0140)	0.0622*** (0.0139)
Top3Share $\times$ Top3_BPM	-0.0590** (0.0227)	-0.0391* (0.0217)
Taxable Salary	1.35×10^{-9} *** (4.43×10^{-10})	1.09×10^{-9} ** (4.72×10^{-10})
MidVet_MinShare		0.2508*** (0.0606)
Team FE	Yes	Yes
Year FE	Yes	Yes
Observations	450	450
R^2	0.551	0.573
Within R^2	0.439	0.467

Marginal Effect of a 1-Point Increase in Top3_BPM on Win_Pct, by Concentration Level

Top3Share Level	Marginal Effect on Win_Pct	Implied Wins per 82-Game Season
40%	0.0532	4.4
60%	0.0414	3.4

Notes: Standard errors clustered by team in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The negative interaction indicates the marginal benefit of improved star quality shrinks as payroll concentration rises. Adding MidVet_MinShare in Column (2) shrinks the interaction from -0.059 to -0.039 (significant only at the 10% level), consistent with part of this pattern operating through reduced reliance on mid-level veterans. Marginal effects computed from Column (1) coefficients.

**Table 8: Production as a Function of Usage and Salary,
High-Salary Veterans (2025-26)**

Dependent Variable:	EPM		OFF_EPM	
	(1) Linear	(2) Quadratic	(3) Linear	(4) Quadratic
Constant	−3.720*** (0.5438)	0.558 (1.492)	−4.400*** (0.3834)	−1.814* (1.057)
USG	0.1144*** (0.0300)	−0.3132** (0.1423)	0.1669*** (0.0211)	−0.0917 (0.1008)
Salary (\$M)	0.0797*** (0.0105)	0.0706*** (0.0107)	0.0579*** (0.0074)	0.0524*** (0.0076)
USG ²		0.0103*** (0.0033)		0.0062*** (0.0024)
Observations	248	248	248	248
R^2	0.408	0.430	0.561	0.573
Adj. R^2	0.404	0.423	0.557	0.568

Notes: IID standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Sample: classified high-salary veterans (at least four years of NBA service, earning more than 13% of the salary cap), 2025-26 season, $N = 248$ across the full comparison population before restricting to residual rankings.

Table 9: Production Relative to Expectation — Jaylen Brown vs. Derrick White (2025-26)

Player	USG (%)	Salary (\$M)	EPM	OFF_EPM	EPM Residual		OFF_EPM Residual	
					Linear	Quadratic	Linear	Quadratic
Derrick White	21.8	28.1	4.4	1.8	+3.39	+3.80	+0.94	+1.18
Jaylen Brown	35.2	53.1	2.2	2.3	−2.34	−3.83	−2.25	−3.15

*Jaylen Brown's Rank Among High-Salary Veterans
($N = 84$), 1 = Most Underperforming*

Specification	Brown's Rank
EPM Residual, Linear	9 of 84
EPM Residual, Quadratic	2 of 84
OFF_EPM Residual, Linear	5 of 84
OFF_EPM Residual, Quadratic	2 of 84

Notes: Residuals are actual minus predicted EPM/OFF_EPM from the corresponding model in Table 8, evaluated for each player. Positive residuals indicate production above what usage rate and salary would predict; negative residuals indicate underperformance relative to that standard. Across all four specifications, Brown ranks among the two-to-nine most-underperforming players (of 84) in the high-salary-veteran comparison group.

Table 10: Multi-Year Robustness — BPM as a Function of Usage, Salary, and Teammate Quality (Linear)

Dependent Variable:	BPM		
	(1) 2022-23	(2) 2023-24	(3) 2024-25
Constant	−4.608*** (0.6058)	−5.380*** (0.6657)	−4.683*** (0.6452)
USG%	0.1372*** (0.0318)	0.1751*** (0.0349)	0.1620*** (0.0329)
Salary (\$M)	0.1306*** (0.0153)	0.1044*** (0.0152)	0.0775*** (0.0126)
Teammate_BPM	0.0927 (0.1186)	0.1124 (0.1180)	0.3198** (0.1092)
Observations	247	256	261
R^2	0.435	0.361	0.330
Adj. R^2	0.428	0.353	0.322

Notes: IID standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Teammate_BPM is the leave-one-out, minutes-weighted average BPM of a player's classified teammates that season, included because (unlike EPM) BPM does not natively adjust for teammate quality.

Table 11: Multi-Year Robustness — BPM as a Function of Usage, Salary, and Teammate Quality (Quadratic)

Dependent Variable:	BPM		
	(1) 2022-23	(2) 2023-24	(3) 2024-25
Constant	1.986 (1.642)	1.282 (1.845)	3.259 (2.080)
USG%	−0.5078*** (0.1531)	−0.4889*** (0.1755)	−0.6023*** (0.1935)
USG% ²	0.0150*** (0.0035)	0.0156*** (0.0040)	0.0174*** (0.0043)
Salary (\$M)	0.1158*** (0.0152)	0.0959*** (0.0150)	0.0708*** (0.0123)
Teammate_BPM	0.0787 (0.1146)	0.1082 (0.1148)	0.3085** (0.1062)
Observations	247	256	261
R^2	0.475	0.397	0.369
Adj. R^2	0.467	0.387	0.359

Notes: IID standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Quadratic specification allows returns to usage to accelerate at the top of the distribution, where a small number of true superstars carry disproportionate offensive loads.

Table 12: Jaylen Brown’s Residual and Rank Among High-Salary Veterans, 2022-23 to 2024-25

Panel A: Residual and Rank (BPM, with Teammate_BPM Control)

Season	Residual (Linear)	Rank (Linear)	Residual (Quad.)	Rank (Quad.)	N High-Sal-Vet
2022-23	−2.34	13 of 84	−3.01	6 of 84	84
2023-24	−2.70	18 of 87	−2.90	12 of 87	87
2024-25	−4.80	2 of 93	−4.79	2 of 93	93

Panel B: Raw BPM and Salary Trend

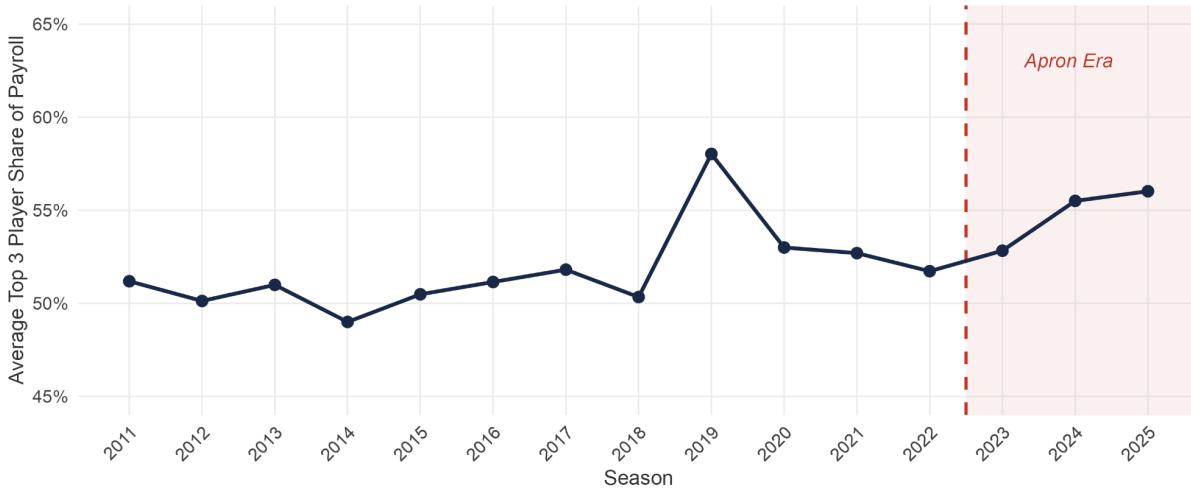
Season	Year	BPM	Salary (\$)
2020-21	2020	2.5	23,735,118
2021-22	2021	2.1	26,758,928
2022-23	2022	1.3	28,741,071
2023-24	2023	0.7	31,830,357
2024-25	2024	0.0	49,205,800

Notes: Panel A: rank is out of the classified high-salary-veteran tier that season (1 = most underperforming). Residuals from the corresponding season’s regression in Table 10 (linear) / Table 11 (quadratic). Panel B: raw, unadjusted BPM and salary, included to show that Brown’s shortfall relative to expectation coincides with a period of declining raw production and sharply rising pay — his BPM fell from 2.5 to 0.0 while his salary grew from \$23.7M to \$49.2M.

Appendix B: Figures

Figure 1: League-Average Payroll Concentration Over Time

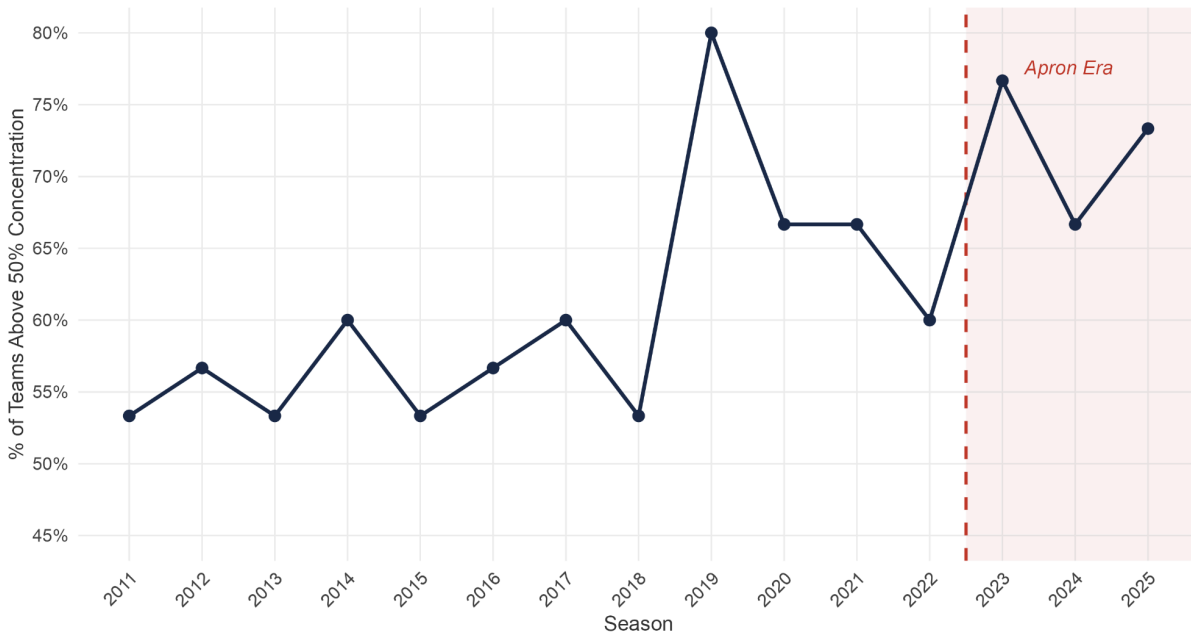
League-wide average share of payroll allocated to top 3 players per season



Notes: Each point represents the league-wide average Top3Share across all 30 NBA teams in that season. The dashed red line marks the introduction of the second apron under the 2023 CBA. The shaded region indicates the apron era (2023-25). Sample period: 2011-2025.

Figure 2: Teams Allocating More Than Half of Payroll to Top 3 Players

Share of teams with Top3Share above 50%, by season

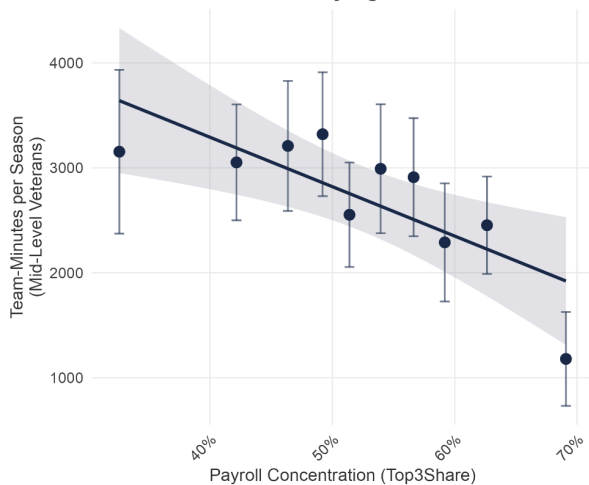


Notes: Each point represents the share of NBA teams allocating more than half of total payroll to their three highest-paid players in that season. The dashed red line marks the introduction of the second apron under the 2023 CBA; the shaded region indicates the apron era (2023-25). Sample period: 2011-2025.

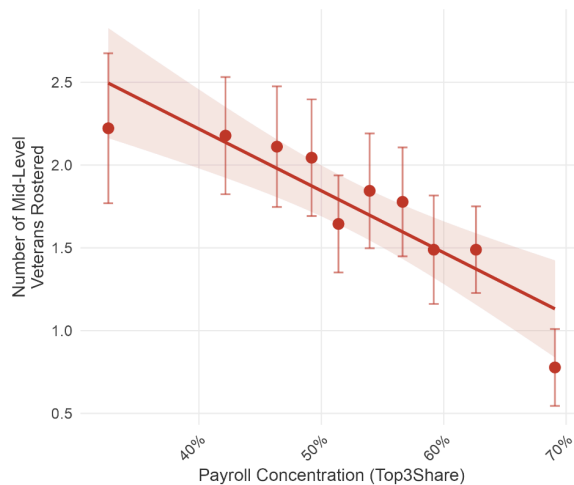
Figure 3: As Payroll Concentration Rises, Teams Reduce Reliance On Mid-Level Veterans

Team-seasons grouped into 10 equal-sized bins by payroll concentration; points show bin averages with 95% confidence intervals

Panel A: Mid-Level Veteran Playing Time



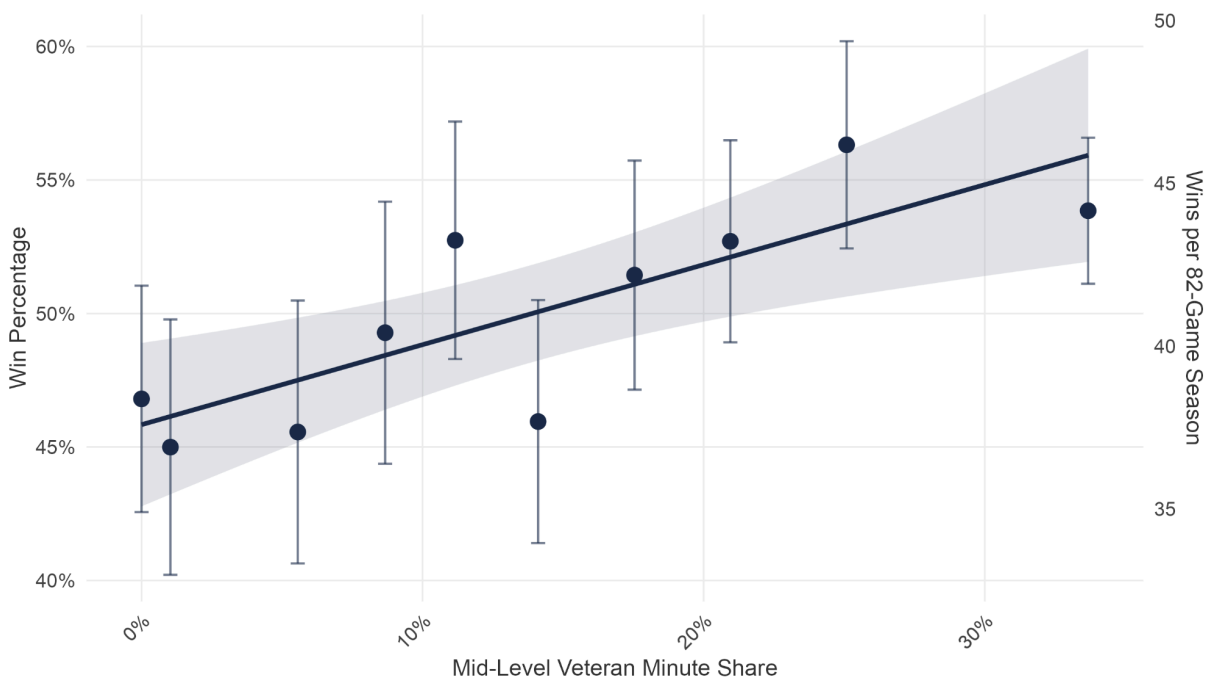
Panel B: Mid-Level Veterans Rostered



Notes: Each point represents the average outcome for team-seasons within that concentration decile ($n \approx 45$ per bin). Shaded bands show the 95% confidence interval of the underlying linear fit. Panel A converts predicted mid-level veteran minute share into team-minutes over an 82-game season (19,680 total team-minutes). Sample: 2011-2025, $N = 450$ team-seasons.

Figure 4: Mid-Level Veteran Minute Share Associated With Winning

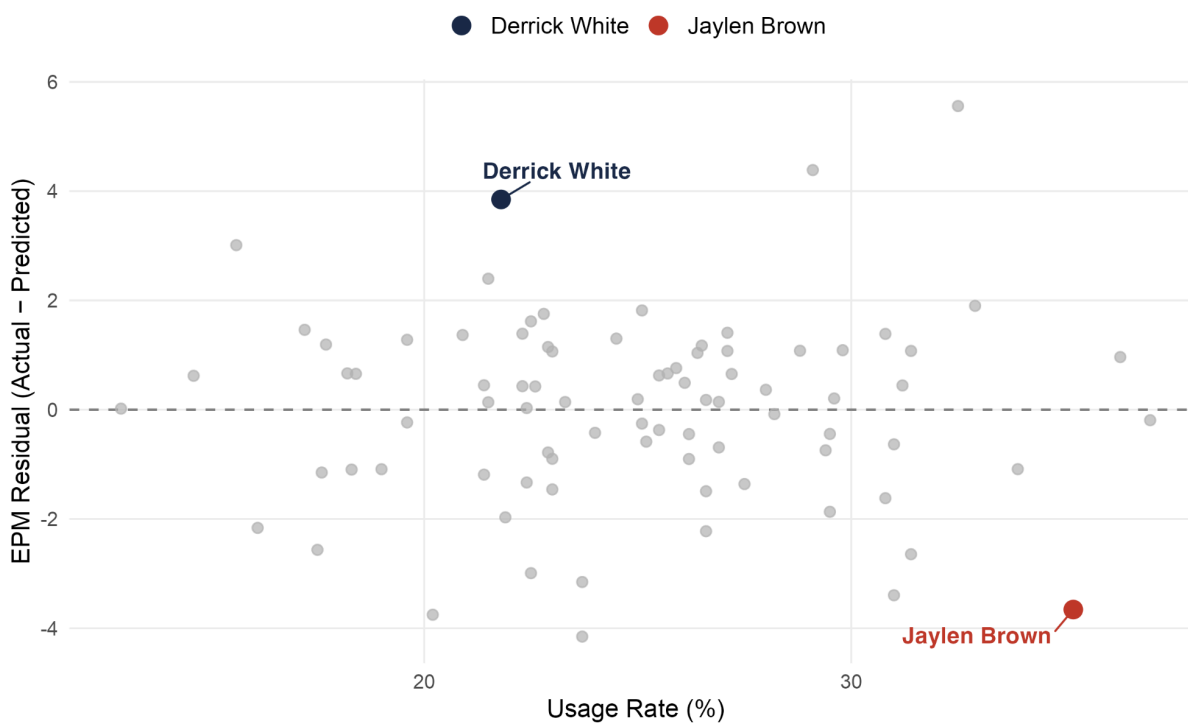
Team-seasons grouped into 10 equal-sized bins by mid-level veteran minute share



Notes: Each point represents the average win percentage for team-seasons within that decile of mid-level veteran minute share ($n \approx 45$ per bin). Shaded band shows the 95% confidence interval of the underlying linear fit. Sample: 2011-2025, $N = 450$ team-seasons.

Figure 5: Production Relative to Expectation, High-Salary Veterans

EPM residual (actual minus predicted, given usage and salary), 2025-26 season



Notes: Each point is a classified high-salary veteran (at least four years of service, earning more than 13% of the salary cap). Residuals are from a quadratic regression of EPM on usage rate and salary. Points above the dashed line outperformed expectation; points below underperformed it.

Figure 5 Interactive Link: https://andrewdaileanes.github.io/figure5_interactive.html